

Prudential Data Quality Report

Profiling and missing-value cleansing, run against the real 50,000-row extract.

Prepared by	Neelam Lahre
Dataset	data_raw/prudential_sg_raw.csv — 50,000 rows x 65 columns
Scripts run	python/01_profile.py -> python/02_clean.py
Guide reference	Solutioning Guide v1.0, Stage 2 and Stage 3
Status	Stages 2-3 complete. Stage 4 (duplicates) not yet started.

50,000	715,507	899	49,101
rows loaded	cells normalised to null	rows quarantined	rows carried to Stage 4

Stage 2 — Profile the Raw Extract

Goal: understand the damage before touching anything.

01_profile.py loaded all 65 columns as text (**dtype=str, keep_default_na=False**) so pandas could not silently hide any defect, then measured blank-likeness column by column against twelve known fake-null spellings, and wrote the result to **logs/profile_before.csv**.

Dirtiest columns (by blank %)

Column	Blank %	Read
claim_decline_reason	92.97%	Structural — only declined claims carry a reason
lapse_date	90.55%	Structural — only lapsed policies have one
rider_premium_before/after_sgd	83.82%	Structural — only riders attached
copay_pct / deductible_sgd	83.2–83.8%	Structural — health rider fields only
ip_plan_tier	78.46%	Structural — Integrated Shield Plan holders only
agent_id / agent_name / agent_tier	47.5–47.7%	Structural — Bancassurance/Digital/Direct have no agent
claim_id / claim_status / assessor_id	40.1–40.4%	Structural — policies with no claim filed

Real defects — mandatory fields with blanks

No data dictionary shipped with this extract, so the mandatory-field list was reconstructed from the guide's own Stage 3.5 critical-field set. These are not structural — every one is a genuine gap:

Field	Blank rows	Blank %
customer_id	322	0.64%
policy_status	272	0.54%
annual_premium_sgd	272	0.54%
distribution_channel	271	0.54%
gender	271	0.54%
product_category	257	0.51%
policy_id	245	0.49%

Cardinality drift confirmed: distribution_channel showed 34 distinct spellings (AGENCY, Bncssurnc, DIGITL, ...) against 5 true channels — the typo/casing problem the guide expects, deferred to Stage 6.

Proved: logs/profile_before.csv exists with 65 rows · 3+ real-defect columns named above · distribution_channel cardinality drift confirmed (34 vs. 5).

Stage 3 — Cleanse Missing Values and Null Tokens

Goal: make “missing” mean one thing — leave null, derive, impute, or quarantine, never impute money or keys.

02_clean.py normalised every fake-null spelling to a true null, derived age from date_of_birth where age itself was missing, imputed four low-risk descriptive fields with their column mode, and quarantined any row missing a business key or financial amount rather than guessing one.

Audit trail (logs/dq_report_stage3.csv)

Check	Records affected	Treatment
null_standardised	715,507	12 fake-null spellings -> 1 true null
age_derived_from_dob	246	Derived
occupation_group_imputed	302	Imputed (mode)
postal_district_imputed	282	Imputed (mode)
residency_status_imputed	299	Imputed (mode)
marital_status_imputed	0	Imputed (mode) — none needed
rows_quarantined_missing	899	Quarantined — critical field missing

Judgment call for data-owner sign-off: the guide's own walkthrough expects roughly 250–400 quarantined rows; this real extract produced 899 — over double. Nothing was deleted: all 899 rows sit in

data_quarantine/q_missing_critical.csv with reason **MISSING_CRITICAL_FIELD**, per the guide's golden rule. Worth raising before Stage 6 locks in category standardisation.

Proved: 715,507 cells now genuinely null · 899 rows quarantined, 49,101 rows continue · logs/dq_report_stage3.csv and data_quarantine/q_missing_critical.csv both exist.

Next: Stage 4 — exact and fuzzy duplicates, run against the 49,101 rows this stage produced.